

ANSWER BOOK

**I AGREE TO ABIDE BY THE SENATE POLICY ON CONDUCT OF ASSIGNMENTS,
MID-TERM EXAM, AND FINAL EXAM**

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Assignment on **Genetic Algorithms**:

Q2. Solution:

Generation 1 - Mutation Process:

i	Population	Site	Mutated Population	Value x_i	Fitness $f(x_i)$
1	001.0000	2	011.0000	3.0000	3.9841
2	010.0000	3	011.0000	3.0000	3.9841
3	011.0100	4	011.1100	3.7500	15.7480
4	100.0000	5	100.0100	3.2500	15.7480
5	101.1000	6	101.1010	5.6250	0.2214
6	110.1000	7	110.1001	6.5625	0.1066

Fitness Sum = **39.7922**; Fitness Average = **6.6320**